

ABHIPRIYO BISWAS

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PROFESSIONAL SUMMARY

Software Engineer with 7+ years of experience spanning **full-stack web development**, **backend/API engineering**, and **mobile application development**, and holds an **MSc in Data Science (CGPA: 7.78)**, completed alongside this engineering career. Career progression from Web Designer to Frontend Developer to Full Stack Developer, with hands-on production experience across **React.js**, **AngularJS**, **React Native**, **Node.js**, **Python**, and **FastAPI**, building and maintaining downstream APIs, state management (Redux), caching (Redis), and secure credential management (GCP Secret Manager) for high-traffic enterprise applications. Additionally skilled in end-to-end **ETL/ELT data pipelines** (Python, PostgreSQL, Apache Airflow, dbt), web automation and scraping (Selenium, BeautifulSoup), **interactive dashboards** (Plotly, Dash, Power BI), and **machine learning models** (Scikit-learn, XGBoost, TensorFlow, PyTorch), including a deep learning-based breast cancer detection system. Comfortable owning work end-to-end — from production debugging and unit testing to data pipeline design and model evaluation — within Agile, cross-functional, and distributed teams.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), TypeScript, SQL, Bash

Frontend: React.js, AngularJS, Angular, React Native, Next.js, Redux, Redux-Thunk, Zustand, HTML5, CSS3, Bootstrap, jQuery, Responsive Design, Component-Based Architecture

Backend & APIs: Node.js, Python, FastAPI, REST API Design & Integration, Layered API Architecture, JSON, HTTP, Redis (Caching)

Web Automation & Scraping: Selenium (JS-Rendered Site Automation), BeautifulSoup, Requests, Fault-Tolerant/Error-Isolated Scraping Architecture, Scheduled Automation (APScheduler/cron)

Data Engineering: ETL/ELT Pipelines, dbt, Apache Airflow, Data Modeling, Data Warehousing, Batch Processing, Data Transformation & Validation, Idempotent Upserts, Pipeline Orchestration

Data Analysis & Visualisation: Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering, Pandas, NumPy, Matplotlib, Plotly, Dash, Power BI (familiar), KPI Design, Reporting

Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Clustering, SMOTE, Cross-Validation, Hyperparameter Tuning, Model Evaluation (ROC-AUC, F1, Precision/Recall), TensorFlow, PyTorch, CNN (Medical Imaging), Transfer Learning

Databases & SQL: PostgreSQL, MySQL, Advanced SQL (Window Functions, CTEs, Joins, Aggregations, Subqueries, Indexing, Query Optimisation)

Cloud & DevOps: GCP (BigQuery, Secret Manager, Cloud Logging), Docker, Git, GitHub, Jenkins (CI/CD)

Testing & Quality: Unit Testing, Jest, Enzyme, React Testing Library, Debugging, Production Issue Resolution, Error Handling, Logging

Big Data & Orchestration: Apache Airflow, Apache Spark (familiar)

Statistical Analysis: Hypothesis Testing, Descriptive Statistics, Distribution Analysis, Resampling Methods

Tools: VS Code, Jupyter Notebook, Overleaf, Linux

Methodologies: Agile, SDLC, Modular Architecture, End-to-End Project Delivery, Cross-functional & Distributed Team Collaboration, Documentation

PROFESSIONAL EXPERIENCE

Career Progression: Web Designer → Frontend Developer → Full Stack Developer

Full Stack Developer / Contractor

Aug 2021 – Dec 2025

TELUS Digital (via Etelligens Technologies)

Remote, Noida, India

- Worked as a Contractor on the **Sapphire team**, contributing to **CASA**, a customer support application used by call center agents to access customer profiles, billing information, service plans, offers, and account dashboards in real time.
- Delivered real-time customer data access for call center agents by developing and enhancing features in a **React.js** and **Node.js** application, integrating secure backend services to retrieve and display offers data from downstream APIs.

- Shipped production features by implementing **Redux** and **Redux-Thunk** for reliable client-side state management of backend API responses.
- Improved backend maintainability and security by collaborating with API teams to implement a **layered API architecture**, where frontend services consumed intermediary APIs that securely accessed and aggregated data from backend systems.
- Reduced frontend payload complexity by building backend services in Node.js/JavaScript that performed data sanitization, returning only relevant data to the frontend.
- Developed custom UI components and tested using Storybook.
- Strengthened code reliability by writing unit tests using **React Testing Library** and Jenkins for frontend, and **Jest** for backend; used **Redis** for caching and **GCP Secret Manager** for secure credential storage.
- Maintained live service availability during active customer calls by monitoring **GCP Cloud Logging** for errors and warnings, and debugging/resolving production issues faced by customer care agents in real time.
- Improved system scalability and deployment independence by participating in the organization's transition from monolithic architecture to **microservices**, decomposing application modules into independently deployable services.
- Improved API response times and system reliability by later joining the **API development team**, designing and maintaining downstream APIs using **Python**, **FastAPI**, and **PostgreSQL** to support data integration, business workflows, and enterprise applications; optimized backend processing logic and database queries.
- Delivered consistent production stability by working within Agile teams using **Git**-based version control, contributing to production releases across customer-facing systems.

Frontend Developer

Dec 2020 – Dec 2025

Etelligens Technologies

Remote, Noida, India

- Maintained and shipped new features for **TravelNet**, a hybrid mobile application, and **Alloodo**, a game-rental client website, through ongoing frontend maintenance and feature enhancement.
- Developed production-grade **React.js** and **AngularJS** applications, consuming backend REST APIs and optimizing data-driven UI performance across 5+ client projects, including **Pyro**, **Local Kings**, and **Huskarma**.
- Delivered consistent, on-schedule feature releases by collaborating with distributed engineering teams in remote environments across Agile sprint cycles.

Mobile Application Developer

Jun 2020 – Dec 2020

Globiz Technology Inc.

Remote, Ludhiana, India

- Member of both the design and development teams; delivered fully responsive, cross-browser-tested, pixel-perfect static website designs from mockups, and pixel-perfect mobile app screens from UI mockups in **React Native**.
- Delivered performant, production-ready mobile experiences by developing hybrid mobile applications using **React Native**, consuming RESTful APIs and integrating structured backend data, including **LocalCart** and **LineManagement** (deployed to the Google Play Store).

Web Designer / Associate Frontend Developer

Aug 2018 – May 2020

Promatics Technologies Pvt. Ltd.

Ludhiana, India

- Promoted from **Web Designer** to **Associate Frontend Developer** within one year in recognition of strong technical performance, rapid skill development, and consistent project contributions.
- Designed responsive website mockups, mobile application UI mockups, and branding assets (logos, banners, marketing graphics) using **Adobe Photoshop** and **Adobe Illustrator**.
- Converted approved UI/UX designs into responsive, cross-browser compatible web interfaces using **HTML5**, **CSS3**, **JavaScript/jQuery**, and **Bootstrap**.
- Collaborated with designers, developers, and project managers to deliver multiple client-facing websites from design through frontend implementation.
- Contributed to frontend feature development, UI enhancements, bug fixes, and production-ready web pages across multiple projects.
- Built and maintained reusable UI components while ensuring responsive design and a consistent user experience across desktop and mobile devices.
- **Selected Portfolio:** UCatch, PortaPrep, Babzer, GPS Colombia (Design).

DATA ENGINEERING & MACHINE LEARNING PROJECTS

Job Listings Aggregator | *Python, Selenium, BeautifulSoup, FastAPI, SQLAlchemy, PostgreSQL, Pandas, Docker* **2026**
[Source Code]

- Built a fault-tolerant web automation pipeline from scratch, aggregating job listings from multiple sources into a searchable REST API and dashboard, using **Selenium** for JS-rendered, dynamic sites and requests/**BeautifulSoup** for static ones, with per-source error isolation ensuring a single failed page or site never breaks the overall pipeline.

- Built a **FastAPI** REST API exposing 6+ combinable filters (skill, location, remote status, experience, source, free-text search) and pagination over the aggregated dataset.
- Prevented duplicate postings across repeated scrapes by implementing hash-based deduplication with a database-level unique constraint.
- Automated hourly refresh via APScheduler/cron, with email digests for newly discovered postings.
- Ensured reliable, repeatable deployment by containerizing the full pipeline with **Docker** (including headless Chromium + chromedriver); validated correctness with 11 automated tests covering the API, filters, and deduplication logic.

CoinPulse — Crypto Market ETL Platform | *Python, PostgreSQL, Apache Airflow, dbt, Docker, REST API*, **2026**
Data Warehousing [Source Code]

- Eliminated manual data collection and processing by designing and deploying a production-grade end-to-end ETL/ELT pipeline, automating daily extraction and transformation of live cryptocurrency market data (CoinGecko REST API) into a PostgreSQL data warehouse.
- Enforced data quality at the transformation layer by writing **dbt** models converting raw ingested data into structured Silver and Gold layer tables, with built-in dbt tests.
- Prevented duplicate records and maintained data integrity across repeated pipeline executions by implementing idempotent upsert logic and primary-key conflict handling.
- Automated pipeline scheduling and workflow orchestration using **Apache Airflow** DAGs within a Dockerized environment.
- Surfaced historical price trends, asset performance KPIs, and market movement insights across 100+ cryptocurrency assets through SQL analytics reports.
- Improved maintainability and reduced code duplication by designing a modular Python architecture with reusable components and centralized configuration management.
- Ensured reliable ingestion and transformation of API data through comprehensive validation checks and error handling mechanisms.

LogPulse — Application Log Analytics Pipeline | *Python, PostgreSQL, Batch Processing, Data Modeling, SQL*, **2026**
Analytics [Source Code]

- Converted unstructured raw log text into clean, analytics-ready relational data by building a scalable batch data pipeline to ingest and process synthetic application logs into structured records in PostgreSQL.
- Ensured clean, analytics-ready datasets by implementing data parsing, field extraction, and multi-stage data validation checks.
- Enabled monitoring of request volume, HTTP status distribution, and application error trends by developing SQL-based analytical reports and KPI dashboards.
- Supported high-volume log data ingestion with built-in data quality enforcement through scalable, engineered ingestion design patterns.
- Enabled efficient querying and reporting workloads by designing normalized relational schemas and indexing strategies.

Credit Card Fraud Detection — ML Data Pipeline | *Python, Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering, Data Preprocessing* [Source Code], **2025**

- Produced clean, model-ready data from a transactional dataset containing over 1M+ records by building end-to-end data preprocessing and feature engineering pipelines.
- Improved training data quality under significant class imbalance by applying data cleaning, normalization, and SMOTE-based resampling techniques.
- Delivered standardized, model-ready datasets with validated feature distributions and schema consistency by developing automated transformation workflows compatible with Scikit-learn and XGBoost estimators.
- Derived actionable insights on transaction risk patterns by applying cross-validation and performance benchmarking.

Interactive Movie Analytics Dashboard | *Python, Pandas, NumPy, Plotly, Dash* [Source Code], **2025**

- Enabled non-technical stakeholders to explore movie revenue, ratings, trends, and performance metrics without writing queries, by building a fully interactive, production-ready dashboard.
- Created dynamic visualizations including KPIs, bar charts, trend analysis, and distribution plots, backed by full data preprocessing (cleaning, deduplication, metric creation) and user-driven filters.

Breast Cancer Detection — MSc Dissertation | *Python, CNN, TensorFlow, PyTorch, Medical Imaging, Statistical Analysis*, **2025 – 2026**

- Achieved strong diagnostic classification performance on histopathological image data by developing a deep learning model applying Convolutional Neural Networks (CNNs) with transfer learning to classify malignant vs. benign tissue samples.
- Ensured clinical-grade reliability by conducting statistical analysis of model outputs, including ROC-AUC, precision/recall, and confusion matrix evaluation.
- Improved minority-class recall under class imbalance by applying data augmentation and resampling strategies.
- Demonstrated end-to-end project ownership by producing detailed research documentation covering methodology, experimental design, results analysis, and conclusions.

EDUCATION

Master of Science in Data Science
Central University of Haryana, Mahendergarh

Jul 2024 – Jun 2026
CGPA: 7.78 — Completed; Degree Certificate Awaited

- Coursework: Data Engineering, Big Data, Database Management Systems (DBMS), Artificial Intelligence, Machine Learning, Statistical Analysis, Data Visualisation.
- Dissertation: Breast Cancer Detection using Deep Learning and Medical Image Classification.

Bachelor of Arts in Computer Science
Lovely Professional University

Dec 2020 – Apr 2024

CERTIFICATIONS

HackerRank Skill Certifications: **Python (Basic)**, **SQL (Basic)**, **SQL (Intermediate)**

ADDITIONAL INFORMATION

Languages: English, Hindi, Punjabi, Bengali
Open to: On-site, Remote, Hybrid positions and Contractor engagements
Target Roles: Python Backend Developer, Full Stack Developer